Balancing chemical equations

Reactants= the chemicals that your start with. These react together.

Products=the chemicals that are made as a result of the chemical reaction.

$C_3H_8 + O_2$ ----------> $H_2O + CO_2$

Equation= a line which shows the reactants and products in a chemical reaction.

There needs to be the same number of each element on each side of the equation.

Some equations are already balanced.

$C + O_2 \longrightarrow CO_2$

C= **C=**

O= **O=**

Some are not

$H_2 + O_2 \longrightarrow H_2O$

H= **H=**

O= **O=**

If we write the equation as

$2H_2 + O_2 \longrightarrow 2H_2O$

H= **H=**

O= **O=**

It now has equal numbers of each type of element on each side of the equation.

We call the big number 2 written in front of the H_2 and H_2O the <u>coefficient.</u>

How to Balance Most Chemical reactions.

1. Write down your given equation.

$$C_3H_8 + O_2 \text{ --> } H_2O + CO_2$$

2. Write down the number of atoms per each element that you have on each side of the equation.

$$C_3H_8 + O_2 \text{ ----------> } H_2O + CO_2$$

C= H= O=	C= H= O=

3. Leave hydrogen and oxygen for last

4. If you have more than one element left to balance: select the element that appears in only a single molecule of reactants and in only a single molecule of products. This means that you will need to balance the carbon atoms first.

5. Add a coefficient to the single carbon atom on the right of the equation to balance it with the 3 carbon atoms on the left of the equation.

$$C_3H_8 + O_2 \text{ ----------> } H_2O + 3CO_2$$

C= H= O=	C= H= O=

- The coefficient 3 in front of carbon on the right side indicates 3 carbon atoms just as the subscript 3 on the left side indicates 3 carbon atoms.
- In a chemical equation, you can change coefficients, but you must never alter the subscripts.

6. Balance the hydrogen atoms next. You have 8 on the left side. So you'll need 8 on the right side.

2. $C_3H_8 + O_2 \text{ ----------> } 4H_2O + 3CO_2$

C=	C=
H=	H=
O=	O=

- On the right side, you now added a 4 as the coefficient because the subscript showed that you already had 2 hydrogen atoms.
- When you multiply the coefficient 4 times by the subscript 2, you end up with 8.
- The other 6 atoms of Oxygen come from $3CO_2$.(3x2=6 atoms of oxygen+ the other 4=10)

7. Balance the oxygen atoms.

- Because you've added coefficients to the molecules on the right side of the equation, the number of oxygen atoms has changed. You now have 4 oxygen atoms in the water molecule and 6 oxygen atoms in the carbon dioxide molecule. That makes a total of 10 oxygen atoms.
- Add a coefficient of 5 to the oxygen molecule on the left side of the equation. You now have 10 oxygen molecules on each side.

$C_3H_8 + 5O_2 \text{ ----------> } 4H_2O + 3CO_2$

C=	C=
H=	H=
O=	O=

Try these

1.

$CH_4 + O_2 \longrightarrow CO_2 + H_2O$

C=	C=
H=	H=
O=	O=

$CH_4 + O_2 \longrightarrow CO_2 + H_2O$

C=	C=
H=	H=
O=	O=

2.

Hydrochloric acid + Calcium metal ⟶ calcium chloride + Hydrogen gas

3.

Solid carbon + Lead oxide ⟶ Lead metal + Carbon dioxide

$C + PbO \longrightarrow$

1. $4\ HCl + O_2 \longrightarrow H_2O + 2\ Cl_2$

2. $C_3H_8 + 5\ O_2 \longrightarrow 3\ CO_2 + 4\ H_2O$

3. $3\ PbO + 2\ NH_3 \longrightarrow 3\ Pb + N_2 + 3\ H_2O$

4. $3\ NO_2 + H_2O \longrightarrow 2\ HNO_3 + NO$

5. $4\ Al + 3\ O_2 \longrightarrow 2\ Al_2O_3$

6. $2\ NO + O_2 \longrightarrow 2\ NO_2$

7. $3\ H_2 + Fe_2O_3 \longrightarrow 2\ Fe + 3\ H_2O$

8. $2\ H_2S + SO_2 \longrightarrow 3\ S + 2\ H_2O$

9. $2\ CH_3OH + 3\ O_2 \longrightarrow 2\ CO_2 + 4\ H_2O$

10. $6\ P_2H_4 \longrightarrow 8\ PH_3 + P_4$

ANSWERS on next page

1. $4\ HCl + O_2 \longrightarrow H_2O + 2\ Cl_2$

2. $C_3H_8 + 5\ O_2 \longrightarrow 3\ CO_2 + 4\ H_2O$

3. $3\ PbO + 2\ NH_3 \longrightarrow 3\ Pb + N_2 + 3\ H_2O$

4. $3\ NO_2 + H_2O \longrightarrow 2\ HNO_3 + NO$

5. $4\ Al + 3\ O_2 \longrightarrow 2\ Al_2O_3$

6. $2\ NO + O_2 \longrightarrow 2\ NO_2$

7. $3\ H_2 + Fe_2O_3 \longrightarrow 2\ Fe + 3\ H_2O$

8. $2\ H_2S + SO_2 \longrightarrow 3\ S + 2\ H_2O$

9. $2\ CH_3OH + 3\ O_2 \longrightarrow 2\ CO_2 + 4\ H_2O$

10. $6\ P_2H_4 \longrightarrow 8\ PH_3 + P_4$

Balancing with radicals

If you have radicals that exist on both sides of the equation, you can treat them as discrete units

Calcium + gas Nitric acid Calcium nitrate + hydrogen

$Ca + HNO_3 \longrightarrow Ca(NO_3)_2 + H_2$

Ca X 1 H X 1 NO_3 X 1	Ca X 1 H X 2 NO_3 X 2

If a coefficient of 2 is put in front of nitric acid the total number of nitrates is now 2 on both sides of the equation.

$Ca + 2\ HNO_3 \longrightarrow Ca(NO_3)_2 + H_2$

Ca X 1 H X 2 NO_3 X 2	Ca X 1 H X 2 NO_3 X 2

Try the ones on the next page

Magnesium + nitric acid → Magnesium nitrate + Hydrogen gas

Nitric acid + Calcium metal → Calcium nitrate + hydrogen gas

Zinc metal + Silver nitrate → Zinc nitrate + Silver metal

Phosphoric acid + Potassium hydroxide → potassium phosphate + water

Zinc metal + Phosphoric acid → zinc phosphate + Hydrogen gas